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A DISTRIBUTED ASYNCHRONOUS RELAXATION ALGORITHM FOR THE ASSIGNMENT PROBLEM

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Abstract

Relaxation methods for optimal network flow problems resemble classical coordinate descent, Jacobi, and Gauss-Seidel methods for solving unconstrained nonlinear optimization problems or systems of nonlinear equations. In their pure form they modify the dual variables (node prices) one at a time using only local node information while aiming to improve the dual cost. They are particularly well suited for distributed implementation on massively parallel machine. For problems with strictly convex arc costs they can be shown to converge even if relaxation at each node is carried out asynchronously with out-of-date price information from neighboring nodes [1].

For problems with linear arc costs relaxation methods have outperformed by a substantial margin the classical primal simplex and primal-dual methods on standard benchmark problems [2], [3]. However in these particular methods it is necessary to change sometimes the prices of several nodes as a group in addition to carrying out pure relaxation steps. As a result global node price information is needed occasionally, and distributed implementation becomes somewhat complicated.

In this paper we describe a distributed algorithm for solving the classical linear cost assignment problem. It employs exclusively pure relaxation steps whereby the prices of sources and sinks are changed individually on the basis of only local (neighbor) node price information. The algorithm can be implemented in an asynchronous (chaotic) manner, and seems quite efficient for problems with a small arc cost range. It has an interesting interpretation as an auction where economic agents compete for resources by making successively higher bids.

1. Assignment by Means of an Auction

Consider N persons wishing to divide among themselves N objects by means of an auction-like process. In the end each object j will be assigned to exactly one person who will then pay a price p_j (yet to be determined) for that object. There is a fixed value a_{ij} that person i associates with object j and each person i wishes to maximize the benefit $a_{ij}-p_j$ from being assigned to object j. Based on this fact a satisfactory assignment will result if the final prices p_j, and, the object j_i assigned to person i, are such that for all i

$$a_{ij_i}-p_{j_i} = \max_j \{a_{ij}-p_j\} \triangleq \pi_i. \tag{1}$$

By using linear programming theory it is possible to show that such an assignment is optimal in the sense that $\sum_{i=1}^N a_{ij_i}$ is maximal over all possible assignments while the prices p_j and "profit margins" π_i given by (1) solve the dual problem

$$\text{minimize} \quad \sum_{i=1}^N \pi_i + \sum_{j=1}^N p_j \tag{2}$$

$$\text{subject to} \quad \pi_i + p_j \geq a_{ij}, \quad \forall\, i,j.$$

Consider now the following process for auctioning the objects. An arbitrary initial price p_j is given to each object. Furthermore initially either no objects are assigned, or else for each person i assigned to an object j_i equation (1) holds. The process proceeds iteratively and terminates when there are no objects left unassigned. At the beginning of each iteration each person knows the price p_j of each object and whether he is assigned to an object. There are two phases in each iteration; the bidding phase, and the assignment phase described below:

Bidding Phase: Each unassigned person i computes the "current cost" of object j

$$c_{ij} = p_j-a_{ij}, \tag{3}$$

and finds a "best" object j* having minimum cost

$$c_{ij^*} = \min_j c_{ij}, \tag{4}$$

and a "second best" object $\bar{j}$ satisfying

$$c_{i\bar{j}} = \min_{j\neq j^*} c_{ij}.$$

Person i then bids up the price of j* by an increment r_i between ε and $(c_{i\bar{j}} - c_{ij^*}) + \varepsilon$, where ε is a positive constant that is fixed throughout the algorithm. The bid

$$b_{ij^*} = p_{j^*} + r_i, \qquad r_i \varepsilon [\varepsilon, c_{i\bar{j}} - c_{ij^*}+\varepsilon] \tag{6}$$

is communicated to an auctioneer. (This choice of r_i guarantees that, if the bid is accepted, object j* will be at most within ε of being best.)

Assignment Phase: For each object j: If there was a bid for object j its price is raised to the highest bid

$$p_j = \max_i b_{ij}, \tag{7}$$

and the new price is anounced to all persons. Furthermore a person i* that made the highest bid is assigned to the object. Person i* as well as the person previously assigned to object j (if any) are informed of the new assignment. If there was no bid for object j its price is left unchanged.

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2. Properties of the Algorithm

A more detailed analysis of the algorithm may be found in the unpublished report [4]. The following may be shown:

a) The algorithm terminates with an assignment which is within $(N-1)\varepsilon$ of being optimal. Therefore if v^* is the value $\sum_{i=1}^{N} a_{ij}$ of an optimal assignment and $\tilde{v}$ is the value of the best nonoptimal assignment, the algorithm terminates with an optimal assignment provided

$$0 < \varepsilon < \frac{v^*-\tilde{v}}{N-1} . \quad (8)$$

In particular, if all a_{ij} are integer a value $\varepsilon < \frac{1}{N-1}$ suffices to guarantee that an optimal assignment will be found. The proof of termination is based on contradiction. If the algorithm were to continue indefinitely, then the prices of some of the objects would be increased to infinity through successive bidding. Meanwhile all unassigned objects would still be at their starting price and therefore at some point would become more attractive than the objects being bidded on. The proof of the bound (8) is based on the fact that at termination the prices p_j and the profit margins

$$\pi_i = \max_j \{a_{ij}-p_j\}$$

satisfy complementary slackness within ε. This can be translated after some calculation into the bound (8).

b) The total number of arithmetic operations needed for termination with $\varepsilon = \frac{1}{N}$ is bounded by $O(bN^3)$ where

$$b = \max_{i,j} a_{ij} - \min_{i,j} a_{ij} , \qquad a_{ij}\text{: integer.}$$

The algorithm is not polynomial, since examples can be constructed showing that the bound $O(bN^3)$ is sharp. However if b is small as for example in the pure matching problem ($a_{ij} = 1$ or $a_{ij} = 0$) the complexity is satisfactory. Limited computational experimentation shows that the algorithm performs very well for small values of b, but can also work very poorly when b is large. There is no theoretical or experimental estimate of the potential speedup of the algorithm through parallelism. It is also unclear whether the algorithm will compare favorably with distributed versions of the Hungarian method or the relaxation method for the assignment problem.

3. Asynchronous Implementation

The auction algorithm given in Section 1 may be described as synchronous since communication of all relevant information follows all updating that takes place in either the bidding or the assignment phases. Furthermore bidding and assignment are carried out simultaneously for all persons and objects respectively. An asynchronous algorithm (see [5] for a general model) results if each unassigned person makes a bid at arbitrary times on the basis of object price information that may be out dated (because of additional bidding of which the person is not informed). Furthermore assignment of objects may be decided even if some potential bidders have not been heard from. We can similarly show the same termination properties as for the synchronous version of the algorithm subject to two conditions.

a) An unassigned person will bid for some object within finite time, and cannot bid twice (i.e. cannot bid for a second object while awaiting for a reply regarding the disposition of an earlier bid for another object).

b) Whenever one or more bids are received that raise the price of an object then, within finite time, that price must be updated, and its value must be communicated (not necessarily simultaneously) to all persons. Furthermore the new bidder assigned to the object must be informed of this fact simultaneously with receiving the new price.

References

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